Samuel Koovely

Building Y27, Office G25
Winterthurerstrasse 190, 8057 Zürich
samuel.koovely@math.uzh.ch

EXPERIENCE

Teaching Assistant — Sep. 2022 - Ongoing
University of Zurich, Switzerland

- Organizer of *Introduction to Machine Learning* — Spring Term 2025
- Organizer of *Markov Models* — Spring Term 2024
- (Head) Assistant of *Complex Networks Theory and Applications* — Fall Terms 2022-2025

Project Intern and Trainee — Nov. 2020 - Nov. 2021
IBM Research Zurich, Switzerland

- Project intern for MSc Thesis [4]
- Further developed the model introduced in [4] as a post-graduate trainee.

Teaching Assistant — Sep. 2017 - Aug. 2019
ETH Zurich, Switzerland

- Assistant of *Mathematik IV: Statistik* (German) — Spring Term 2019
- Assistant of *Stochastics* (German) — Fall Term 2018
- Assistant of *Probability and Statistics* — Fall Term 2017

EDUCATION

PhD studies in Mathematics — Aug. 2022 - ongoing
University of Zurich, Institute of Mathematics, Institute of Mathematical Modeling and Machine Learning

BSc and MSc studies in Mathematics — Sep. 2015 - Jul. 2021
ETH Zurich, Department of Mathematics

SKILLS

Languages: Italian (Native tongue), English (proficient), French (good), German (good)
Programming Skills: Python (good), R (intermediate), Matlab (intermediate), C++ (basic), Wolfram Language (Basic)

SEMINARS & CONFERENCES

Entropy-Informed Structural Changes Detection, NetSci 2025 — Jun. 2025
What is... Free Convolution?, ZGSM Graduate Colloquium, ETH/UZH — Mai 2025
Information-theoretic framework for the dynamics of temporal networks, NetSci 2024 — Jun. 2024
Random Walks and Community Detection, Complex Networks: Theory, Methods, and Applications - 7th ed. — May 2023

SERVICES TO COMMUNITY

Oragnization: ZGSM Graduate Colloquium, ETH/UZH — Jul. 2025-ongoing
Reviewer for Journals: Physical Review E (2025)
Reviewer for Conferences: NetSci (2025)
Examiner of candidates high-school teacher in mathematics, Kantonsschule Rämibühl MNG — Oct. 2024

Committee member for professorial appointments, UZH 2023-2025

PREPRINTS & PUBLICATIONS

[1] Samuel Koovely. "Generating Temporal Networks with the Ascona Model". In: *https://arxiv.org/pdf/2512.16972* (2025).

[2] Samuel Koovely and Alexandre Bovet. "Entropy-Informed Detection of Change Points in Dynamic Networks". In: *Manuscript in Preparation* (2026).

[3] Samuel Koovely and Alexandre Bovet. "Evolution of Conditional Entropy for Diffusion Dynamics on Graphs". In: *https://arxiv.org/pdf/2510.19441* (2025).

THESES & PROJECTS

[4] Samuel Koovely. "A mathematical framework for COMIC-Tree: an undirected graphical model for T-cell receptors specificity". MSc Thesis. 2021.

[5] Samuel Koovely. "Overview and empirical evaluation of some variations of the PC-algorithm". Semester Research Project. 2020.

[6] Samuel Koovely. "Introduction to Riemann surfaces and covering spaces". BSc Thesis. 2019.